

Backtesting ICES Category 1 Advice: Historical, Current and Future

Constant F_{MSY} and harvest-control-rule counterfactuals for six Northeast Atlantic stocks

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18 August 2026 · Working paper — draft, not for submission

Abstract

Realised stock histories mix the properties of scientific advice with the degree to which that advice was followed. We backtest ICES Category 1 MSY advice for six Northeast Atlantic stocks, comparing the latest WGCSE assessment with two perfect-information counterfactuals on the same Operating Model biology: open-loop fishing at constant ICES SAG F_{MSY} (F_{target}), and the ICES hockey-stick harvest control rule (Advice rule) with true Operating Model SSB and no implementation error. Results are organised as Historical, Current and Future. Both windows use the bh3 Beverton–Holt scenario (R_0 prior from a hockey-stick); ICES harvest-control-rule parameters remain the SAG values attached to bh1 (unconstrained bevholtSV, estimated steepness). All six stocks sit below $MSY B_{trigger}$ in the terminal year. Constant F_{target} rebuilds Irish Sea whiting and Celtic Sea cod well above the trigger; Irish Sea cod ends lower than history because realised F is already far below F_{target} . Celtic Sea whiting never reaches ICES $MSY B_{trigger}$ within 20 years, yet reaches Operating Model B_{MSY} in 16–18 years. Remaining stock–recruitment scenarios and an advised-catch trajectory are forthcoming.

Keywords: ICES advice; harvest control rule; backtest; F_{MSY} ; stock–recruitment; rebuilding.

1 Introduction

Fisheries management advice is judged, in practice, against a single realised history: the sequence of assessed stock states and catches that actually occurred. That history is the joint product of the scientific advice rule and the extent to which managers and fleets implemented it. A stock that is depleted at the end of a management period may have been driven there by a poorly performing harvest control rule (HCR), by systematic departure from the advised catch, or by some combination of the two. Distinguishing these explanations matters for where corrective effort should be

directed. Standard retrospective and hindcast diagnostics applied to stock assessments do not, by themselves, separate advice quality from advice compliance.

The intended design of this study (ICES, 2023) is a backtest of ICES Category 1 MSY advice: reconstruct the perception of stock status in each historical year, apply the advice rule then in force, and project the Operating Model under perfect implementation of the resulting catch advice. Comparing that counterfactual with realised history isolates implementation from the advice rule. Performance is summarised in three linked components used throughout: **Historical** (the hindcast window), **Current** (terminal status), and **Future** (a forward projection from each terminal state). Every metric is reported for the realised trajectory and for the counterfactuals, together with their difference.

This draft implements a restricted version of that design. Operating models are conditioned on the latest Working Group for the Celtic Seas Ecoregion (WGCSE) age-structured assessments. Two harvest policies are then applied with perfect information on the Operating Model: a simple open-loop projection at constant ICES SAG F_{MSY} from 2006 (F_{target} ; no feedback), and a closed-loop MSE with feedback control from 2005 (Advice rule: the ICES hockey-stick revises F from SSB each year). The Advice rule uses true Operating Model SSB (`err = NULL`), zero implementation error, and a one-year advice lag. It is not a contemporaneous peel of historical ICES catch advice, and it is not an estimation-error management strategy evaluation. An advised-catch trajectory that would measure TAC overshoot, override, or mixed-fishery constraint remains forthcoming.

Six Category 1 stocks are retained: Irish Sea cod (*Gadus morhua*, `cod.27.7a`), Irish Sea whiting (*Merlangius merlangus*, `whg.27.7a`), Celtic Sea cod (`cod.27.7e-k`), Celtic Sea whiting (`whg.27.7b-ce-k`), Celtic Sea haddock (*Melanogrammus aeglefinus*, `had.27.7b-k`), and Northeast Atlantic mackerel (*Scomber scombrus*, `mac.27.nea`). Northeast Atlantic mackerel is a special case: a recent benchmark involved a substantial change in the assessment, so the backtest conflates following the advice with robustness of that advice to the assessment change. It is retained for completeness and will be removed from the final report. Advice follows the ICES MSY approach except for mackerel, where the Coastal States MSY rule applies. The stocks span a range of terminal status, exploitation, and reference-point conventions, which is the point of the common Historical / Current / Future template.

2 Methods

2.1 Case-study stocks

The six stocks are listed in Table 1. Operating Model terminal years follow the cleaned WGCSE FLStock objects (Irish Sea cod 2023; Irish Sea whiting 2021; Celtic Sea cod and haddock 2025; Celtic Sea whiting and mackerel 2024). Irish Sea whiting has the earliest terminal year. This is a simulation experiment, not a statutory re-assessment.

Table 1: Case-study stocks. Advice basis is the ICES MSY approach except for Northeast Atlantic mackerel (Coastal States MSY rule). Terminal year is that of the WGCSE FLStock used as the Operating Model.

Stock	SID	ICES area	Assessment	End
Irish Sea cod	<code>cod.27.7a</code>	7.a	Age-structured analytical	2023
Irish Sea whiting	<code>whg.27.7a</code>	7.a	SAM	2021
Celtic Sea cod	<code>cod.27.7e-k</code>	7.e-k	SAM	2025
Celtic Sea whiting	<code>whg.27.7b-ce-k</code>	7.b-c, 7.e-k	SAM	2024
Celtic Sea haddock	<code>had.27.7b-k</code>	7.b-k	SAM	2025
NE Atlantic mackerel	<code>mac.27.nea</code>	1-8, 14, 9.a	Integrated age-structured	2024

2.2 Operating Models and stock-recruitment

Operating Models were conditioned on cleaned WGCSE FLStock biology: time-varying weights, natural mortality, maturity and selectivity as assessed. Six stock-recruitment relationship (SRR) scenarios share that biology:

- `bh1` — unconstrained Beverton–Holt (Beverton and Holt, 1957) fitted with `icesdata::eq1(..., model = "bevholtSV")`. Steepness is estimated, not fixed at 1.
- `bh2` — the same Beverton–Holt form with a FishLife steepness prior.
- `bh3` — Beverton–Holt with an R_0 prior taken from a hockey-stick fit.
- `rk1–rk3` — Ricker (Ricker, 1954) analogues of the three Beverton–Holt scenarios.

This paper reports one scenario at a time, not the full Operating Model set. Historical trajectories (open- and closed-loop histories) and the 20-year Future window are `bh3`. ICES HCR parameters stay at published SAG values on eqs (`eq1s[["bh1"]]`). Operating Model evaluation reference points are `bh3`. Other scenarios are forthcoming.

Recruitment deviations on Historical are that scenario’s `srResiduals(eq)`, replayed once. The iteration dimension is 1: the run is deterministic, not a median of a stochastic ensemble. The 20-year Future window uses the same `bh3` Operating Model SRR with `fwdWindow`, FLBRP mean biology, and recruitment deviations of 1 (mean recruitment). Advice-rule Future starts from the `bh3` closed-loop terminal state. It is longer than the ICES 1–2 year advice forecast and is not a stochastic forecast.

A segmented-regression (hockey-stick) form (Mesnil and Rochet, 2010) is used only as the ICES HCR on the advice side, and as the source of the `bh3` R_0 prior. It is not treated as Operating Model “truth”.

2.3 Harvest policies

Three trajectories share the same Operating Model biology; only the harvest policy differs. Open loop is a simple projection at constant F_{target} . Closed loop is an MSE with feedback control: the ICES advice rule revises F from SSB each year.

- **Historical** — the latest WGCSE assessment (FLStock).
- F_{target} — **open loop**: simple projection; fwd at constant ICES SAG F_{MSY} from 2006 to the terminal year, with that scenario's SRR residuals. No feedback: F does not depend on estimated or true SSB. This is not the time-varying yearly advice-sheet F .
- **Advice rule (AR)** — **closed loop**: MSE with feedback control; hcrICES from 2005 to the terminal year. The hockey-stick sets $F = F_{\text{target}}$ when $\text{SSB} \geq \text{MSY} B_{\text{trigger}}$ and reduces F towards B_{lim} when SSB is below the trigger, so F depends on biomass each year. A TAC is taken from a short-term fwd and applied as Operating Model catch. Settings are `err = NULL` (true Operating Model SSB), `implErr = 0`, and a one-year advice lag. The shortcut estimation-error branch is off.

The open-loop run is the no-feedback benchmark that isolates “fish at F_{MSY} ” from “follow the ICES HCR”. The two can disagree: Irish Sea cod historical F was about $0.07 \times F_{\text{target}}$, so fishing at F_{target} lowers SSB relative to history; Celtic Sea stocks often sit below $\text{MSY} B_{\text{trigger}}$, so the HCR fishes well below F_{target} .

ICES F_{target} , B_{lim} and $\text{MSY} B_{\text{trigger}}$ are taken from SAG (`addBenchmarks on eq1s[["bh1"]]`) and are the HCR inputs (`hcrParams(eqs)`). They are held fixed across SRR scenarios: in reality ICES does not re-derive advice parameters according to which SRR happens to be true. Operating Model F_{MSY} , B_{MSY} and R_0 used for shading and evaluation are `refpts` from `eq1s[["bh3"]]`. Figure lines for B_{lim} , $\text{MSY} B_{\text{trigger}}$ and F_{target} remain the ICES SAG values on `eqs`; trajectories and Operating Model shading are the simulated `bh3` scenario.

2.4 ICES versus Operating Model reference points

Two roles for reference points are kept distinct. ICES F_{target} , B_{lim} and $\text{MSY} B_{\text{trigger}}$ are **operational control points** of the advice rule: they set F each year and stay at published SAG values. Operating Model B_{MSY} and F_{MSY} (`bh3` equilibrium) and years to those levels are **performance metrics**: they score whether the CFP/MSY objective is met on the assumed truth. Reaching ICES $\text{MSY} B_{\text{trigger}}$ does not imply Operating Model B_{MSY} .

[Winker et al. \(2026\)](#) make the same split. The CFP objective is to restore stocks above biomass levels that produce MSY, but the ICES MSY framework has no biomass target: advice is built on F_{MSY} (EqSim, then capped by $F_{P.05}$), B_{lim} , and $\text{MSY} B_{\text{trigger}}$. B_{MSY} is an implicit EqSim output — SSB when fishing at F_{MSY} — and is not reported. $\text{MSY} B_{\text{trigger}}$ is “an operationalised management threshold that triggers a cautious reduction in fishing mortality”, not a target. In practice it is usually $B_{\text{pa}} (\approx 1.4 B_{\text{lim}})$, not the 5th percentile of B_{MSY} ([Winker et al., 2026, 2025](#)). Using the trigger as a surrogate for B_{MSY} therefore classifies stocks as “at MSY” while they are still rebuilding: [Winker et al. \(2026\)](#) find $\text{MSY} B_{\text{trigger}}$ is on average about 60% of B_{MSY} .

The hockey-stick (segmented regression) is how ICES usually derives those control points, not a biological production hypothesis. The SRR is fitted after the assessment; B_{lim} is typically the hockey-stick breakpoint (recruitment impairment), and $\text{MSY} B_{\text{trigger}}$ is then a multiplier of B_{lim} .

EqSim at WKMSYREF almost exclusively used segmented regression (Winker et al., 2026). After the breakpoint, expected recruitment is a constant, so the production function is truncated: for many stocks the breakpoint sits at or near B_{MSY} , which would make a limit essentially equal to the target. This paper therefore does not treat the ICES hockey-stick as Operating Model truth. The HCR shape remains that hockey-stick; the Operating Model SRR is Beverton–Holt (bh3, R_0 prior from the hockey-stick breakpoint — the same BH sensitivity Winker et al. (2026) used when ICES does not supply B_{MSY}).

The backtest therefore needs an Operating Model and a closed-loop run. ICES reference points are not a property of one internally consistent population. An Operating Model conditioned on the WGCSE FLStock gives B_{MSY} and F_{MSY} from the same SRR used to project Historical and Future, so years-to- B_{MSY} is comparable with the hindcast. EqSim as used in ICES benchmarks is largely an open-loop simple projection at constant F : it does not simulate the feedback control loop of MSE (data, advice lag, HCR, catch implementation). Winker et al. (2025) showed that EqSim F_{MSY} is not precautionary relative to a shortcut MSE on an Operating Model that retains assessment structure and advice lags. That is why this study conditions an Operating Model, keeps published ICES HCR parameters as control points, evaluates rebuild against Operating Model MSY performance metrics, and contrasts a simple open-loop projection at F_{target} with a closed-loop MSE with feedback.

For the Irish and Celtic Sea gadoids, $MSY B_{trigger}$ is the ICES SAG $MSY B_{trigger}$. For Northeast Atlantic mackerel that SAG field is not on the same published basis: it is a derived SAG $MSY B_{trigger}$, equal to B_{pa} in the latest SAG row, rather than the long-term management plan / B_{lim} convention on the ICES advice sheet. Mackerel ratios in the tables are therefore not comparable, as published quantities, with the gadoid ratios.

2.5 Historical, Current, Future

Historical is the window from 2005 (Advice rule) or 2006 (F_{target}) to the terminal year, shown as trajectories of F , SSB, recruitment and catch (Figure 1 and the remaining Historical figures). The F_{target} line is ICES F_{target} from SAG; F shading is Operating Model F_{MSY} (bh3). SSB is shaded below Operating Model B_{MSY} (bh3).

Current is terminal-year status: $SSB/MSY B_{trigger}$, absolute $\bar{F}$, and $\bar{F}/F_{target}$. Table 2 reports Historical, F_{target} and Advice rule, with Δ defined as Advice-rule minus Historical $SSB/MSY B_{trigger}$. Under F_{target} , $\bar{F}/F_{target}$ is 1 by construction.

Future is a 20-year HCR projection from each terminal state on the bh3 Operating Model, mean biology and mean recruitment. Both the Historical and Advice-rule series use the ICES HCR after the terminal year; differences are starting state, not a change of rule. Rebuild time is the first projected year in which SSB reaches ICES $MSY B_{trigger}$ or Operating Model B_{MSY} . Already above is recorded as 0 years; not reached within 20 years is recorded as missing.

Software: FLCore, FLBRP, FLasher, FLBacktest (hcrICES), icesdata, and ggplotFL.

3 Results

3.1 Terminal status

All six stocks are below $MSYB_{\text{trigger}}$ in the terminal assessment year (Table 2). Realised $\bar{F}$ exceeds ICES F_{target} except in Irish Sea cod. Constant F_{target} rebuilds Irish Sea whiting ($SSB/MSYB_{\text{trigger}} = 4.94$) and Celtic Sea cod (4.22) to several times the trigger. The Advice rule sets $F = F_{\text{target}}$ once SSB is above the trigger (about a one-year lag); the useful contrast is how far each stock sits below the trigger, and how much SSB the Advice rule gains or loses relative to history (Δ in Table 2).

Irish Sea whiting ($\Delta = 4.45$) and Celtic Sea cod ($\Delta = 4.21$) are the large Advice-rule gains. Celtic Sea haddock ends above the trigger under both counterfactuals ($\Delta = 0.58$). Celtic Sea whiting improves modestly ($\Delta = 0.27$) and remains below the trigger. Mackerel $SSB/MSYB_{\text{trigger}}$ is unchanged at 0.77 ($\Delta = 0.00$); the Advice rule reduces F ($F/F_{\text{target}} = 0.91$) because SSB is still below the trigger. Irish Sea cod is the exception: both counterfactuals lower terminal SSB relative to history ($\Delta = -0.21$).

Table 2: Terminal-year SSB relative to ICES $MSYB_{\text{trigger}}$, absolute $\bar{F}$, and $\bar{F}$ relative to ICES F_{target} (SAG FMSY on eqs). Point values from a deterministic bh3 Historical run. Under F_{target} , F/F_{target} is 1 by construction. Δ is Advice-rule minus Historical $SSB/MSYB_{\text{trigger}}$. Northeast Atlantic mackerel $MSYB_{\text{trigger}}$ is the SAG field (equal to B_{pa} in the latest SAG row), not the LTMP/ B_{lim} convention on the ICES advice sheet.

Stock	Historical			F_{target}			Advice rule			AR – Hist
	SSB/B_{trig}	F	F/F_{target}	SSB/B_{trig}	F	F/F_{target}	SSB/B_{trig}	F	F/F_{target}	Δ
Irish Sea cod	0.54	0.01	0.07	0.27	0.17	1	0.33	0.06	0.37	−0.21
Irish Sea whiting	0.51	1.10	5.25	4.94	0.21	1	4.95	0.21	1.00	4.45
Celtic Sea cod	0.02	1.29	4.44	4.22	0.29	1	4.23	0.29	1.00	4.21
Celtic Sea whiting	0.22	0.53	1.42	0.40	0.37	1	0.49	0.24	0.63	0.27
Celtic Sea haddock	0.63	0.63	1.78	1.21	0.35	1	1.21	0.35	1.00	0.58
NE Atlantic mackerel	0.77	0.27	1.39	0.77	0.19	1	0.77	0.17	0.91	0.00

3.2 Irish Sea cod

Historical $\bar{F}$ is already well below F_{target} ($F/F_{\text{target}} \approx 0.07$), so fishing at F_{target} lowers terminal SSB relative to history (Table 2: F_{target} and Advice-rule $SSB/MSYB_{\text{trigger}}$ both below the Historical ratio). That is the opposite of the other depleted gadoids, where the Advice rule would have cut F (Figure 1).

Two mechanisms can produce a realised F this far below the ICES target, and this backtest does not yet separate them. One is mixed-fishery constraint: Irish Sea cod has been treated as a recovery / choke stock, so catch can sit well below a single-stock TAC even when the advice rule would allow F_{target} . The other is vintage: ICES F_{target} is the SAG FMSY used as an HCR input, whereas Operating Model F_{MSY} (bh3 shading) is a different equilibrium quantity — for this stock the Operating Model value is higher than ICES F_{target} , yet both are far above historical $\bar{F}$. An advised-catch

trajectory would distinguish a choke/recovery TAC from an Operating Model / F_{target} mismatch; it is not in this draft.

On the bh3 Future run, Irish Sea cod reaches Operating Model B_{MSY} before ICES $MSY B_{\text{trigger}}$ (6 versus 8 years from history; 7 versus 8 years from the Advice-rule start; Table 3, Figure 3). That ordering occurs because B_{MSY} sits below ICES $MSY B_{\text{trigger}}$ on this scenario. Rebuilding to the advice threshold and rebuilding to MSY biomass are therefore not the same target.

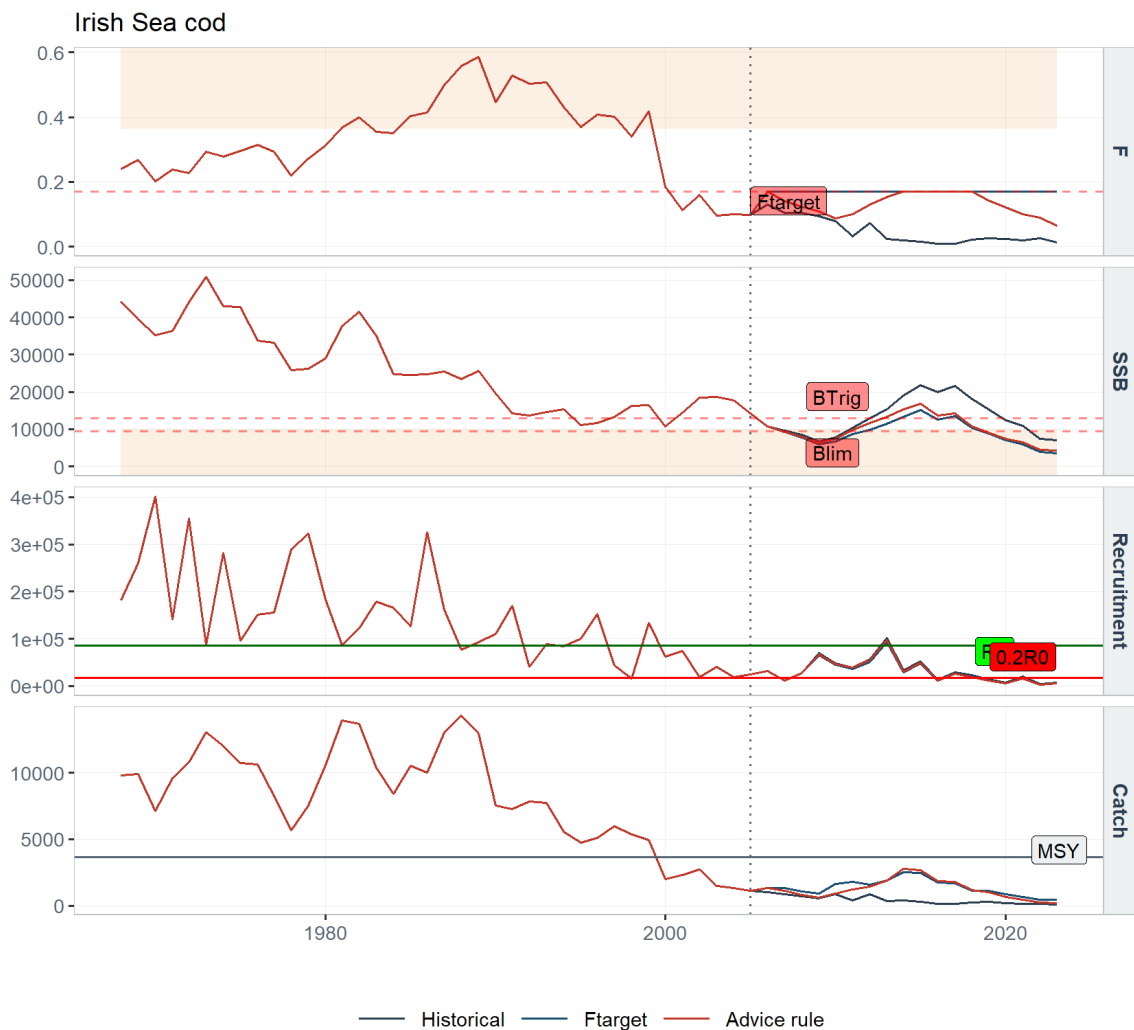


Figure 1: Irish Sea cod Historical: latest WGCSE assessment (Historical), constant ICES F_{target} (F_{target}), and the ICES Advice rule. Trajectories are SRR scenario bh3. B_{lim} and $MSY B_{\text{trigger}}$ are ICES values on eqs. The F_{target} line is ICES F_{target} ; F shading is Operating Model F_{MSY} (bh3). SSB is shaded below Operating Model B_{MSY} (bh3).

3.3 Celtic Sea whiting: $MSY B_{\text{trigger}}$ versus B_{MSY}

Under the bh3 Future run, Celtic Sea whiting never reaches ICES $MSY B_{\text{trigger}}$ in 20 years from either start, but it does reach Operating Model B_{MSY} in 18 years from history and 16 years from the Advice-rule start (Table 3, Figure 4). ICES $MSY B_{\text{trigger}}$ sits materially above Operating Model

B_{MSY} on this scenario, so the advice threshold is not a proxy for MSY biomass. A 20-year mean-recruitment HCR can look like a successful rebuild against the Operating Model yardstick while remaining “below trigger” on the ICES scale.

On the bh3 Historical run, constant F_{target} raises terminal $SSB/MSY B_{trigger}$ only from 0.22 to 0.40, and the Advice rule to 0.49 (Table 2, Figure 2). Realised $F/F_{target} = 1.42$, so the HCR does cut F , but not by enough — given this SRR and these residuals — to put the stock on the ICES trigger within the historical window.

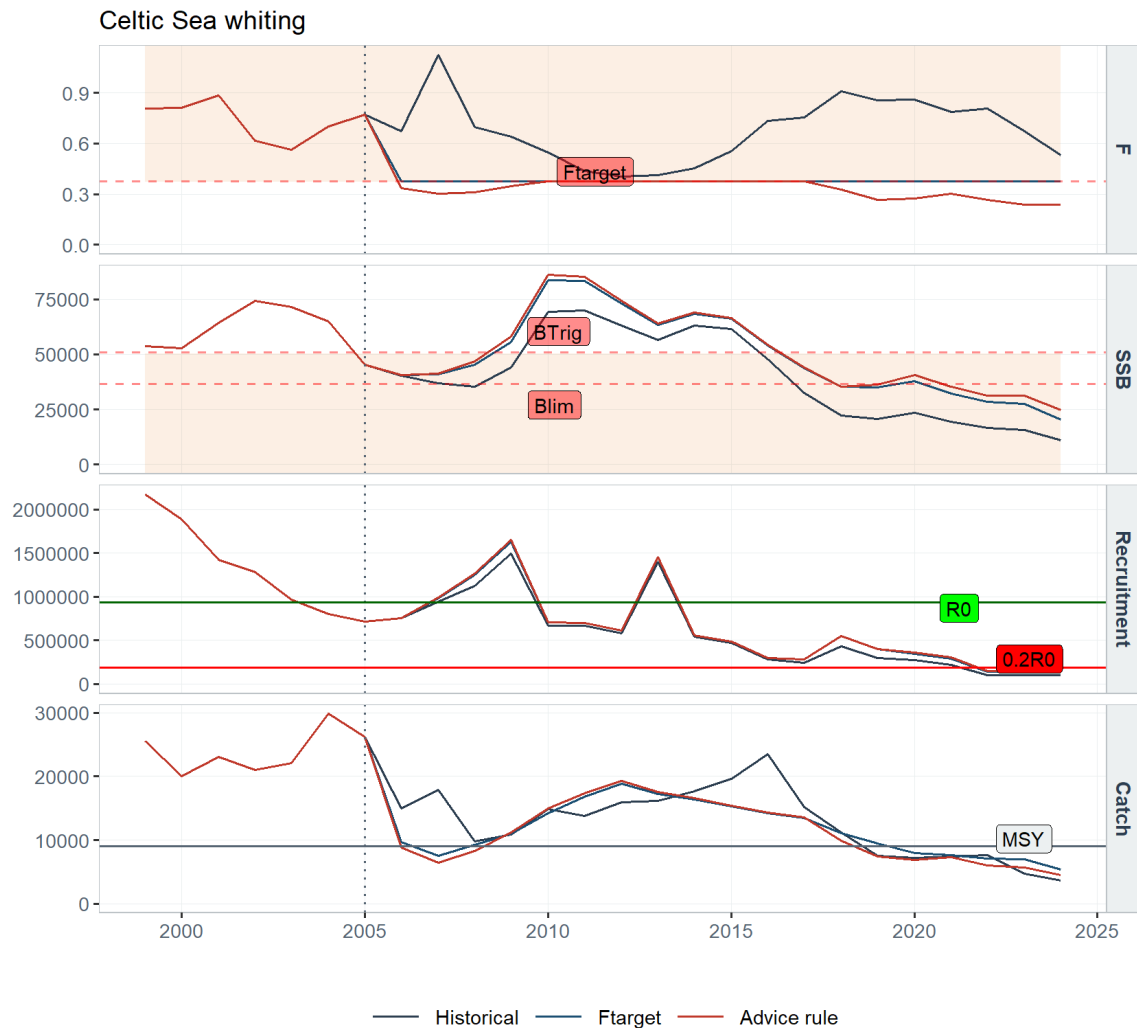


Figure 2: Celtic Sea whiting Historical (bh3). Line and shading conventions as in Figure 1.

3.4 Rebuild times

Table 3 reports years from the terminal year until SSB reaches ICES $MSY B_{trigger}$ and Operating Model B_{MSY} under the ICES HCR (bh3, mean recruitment). Both series use the same HCR after the terminal year.

Irish Sea whiting reaches $MSY B_{trigger}$ in 2 years from history and B_{MSY} in 7; the Advice-rule start is

already above both (0 years). Mackerel is already above B_{MSY} while still below the trigger (3 years from either start) — the inverse of Celtic Sea whiting, and a direct consequence of the mackerel SAG trigger sitting above Operating Model B_{MSY} (Section 2.4).

Celtic Sea cod reaches ICES $MSY B_{trigger}$ in 9 years from the Historical terminal state; Operating Model B_{MSY} is not reached in 20 years from either start (Figure 5). The Advice-rule start is already above the trigger (0 years) but still short of Operating Model B_{MSY} . Celtic Sea haddock reaches the trigger in 4 years from history (already above from the Advice-rule start) and does not reach Operating Model B_{MSY} in 20 years from either start.

Table 3: Years from the terminal year until SSB reaches ICES $MSY B_{trigger}$ and Operating Model B_{MSY} (bh3). Deterministic run (iteration dimension 1), not a median. 0 if already above; – if not within 20 years. Both series use the ICES HCR after the terminal year. Future trajectories are in Figures 3, 4 and 5.

Stock	Historical		Advice rule	
	$MSY B_{trigger}$	B_{MSY}	$MSY B_{trigger}$	B_{MSY}
Irish Sea cod	8	6	8	7
Irish Sea whiting	2	7	0	0
Celtic Sea cod	9	–	0	–
Celtic Sea whiting	–	18	–	16
Celtic Sea haddock	4	–	0	–
NE Atlantic mackerel	3	0	3	0

Historical and Future plots for Irish Sea whiting, Celtic Sea haddock and Northeast Atlantic mackerel are produced by the same 06.0_report.Rmd chunks (06.0_report-backtest-closed-hcr-2,5,6.png and 06.0_report-traj-2,5,6.png).

4 Discussion

4.1 Implementation versus the advice rule

The design question is whether poor terminal status is an implementation failure, an advice-rule failure, or both. This draft does not yet answer that question with an advised-catch counterfactual. What it does answer is narrower and still useful.

F_{target} versus Historical asks what constant ICES F_{target} would have done, given the Operating Model biology and replayed recruitment residuals. Advice rule versus F_{target} isolates the hockey-stick: F is reduced while SSB is below $MSY B_{trigger}$, then returns to F_{target} with a one-year lag. Advice rule versus Historical mixes that HCR with whatever caused realised F to depart from F_{target} — TAC overshoot, mixed-fishery constraint, assessment revision, or a mismatch between SAG F_{target} and Operating Model F_{MSY} .

For Celtic Sea cod, Irish Sea whiting and Celtic Sea haddock, realised F was above F_{target} and the Advice rule would have left the stock at or above the trigger by the terminal year (Table 2). That is the pattern in which following the rule would have changed Current. For Celtic Sea whiting,

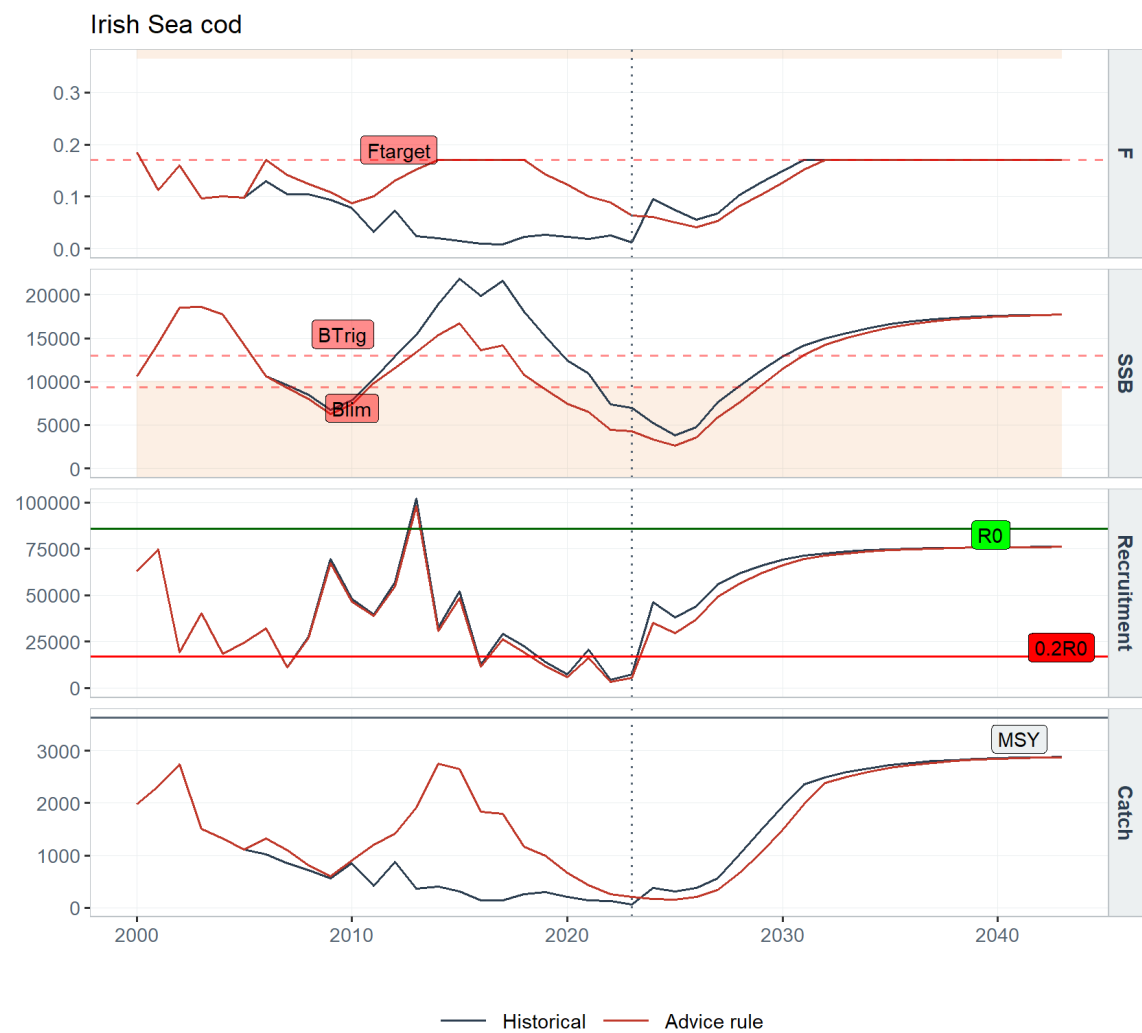


Figure 3: Irish Sea cod Future: Historical and Advice rule through the historical window and a twenty-year HCR projection (bh3 Operating Model, mean biology, mean recruitment). Vertical line is the terminal year. Reference-point conventions as in Figure 1.



Figure 4: Celtic Sea whiting Future (bh3). ICES $MSY B_{trigger}$ is not reached in 20 years from either start; Operating Model B_{MSY} is reached in 16–18 years.

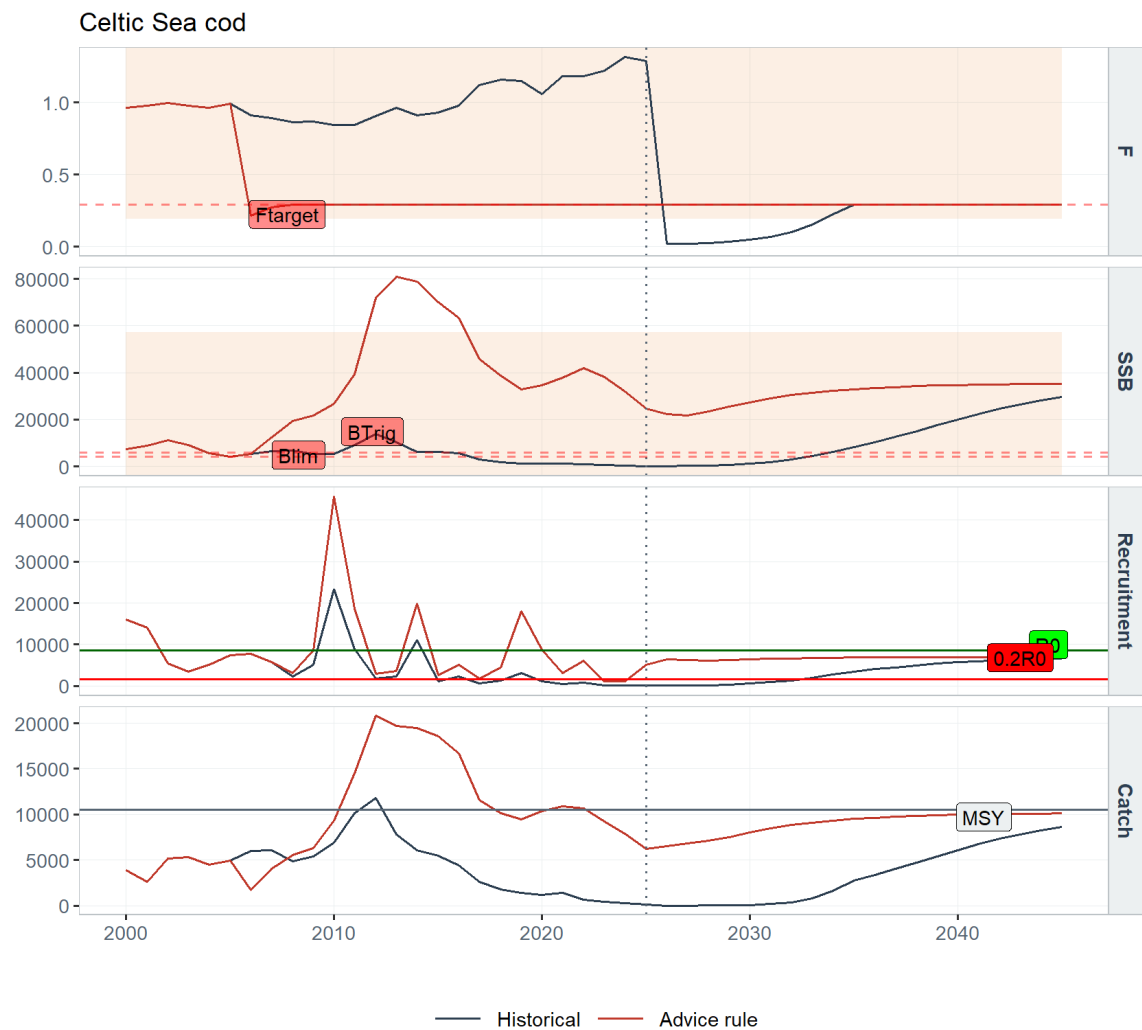


Figure 5: Celtic Sea cod Future (bh3). ICES $MSY B_{trigger}$ is reached in 9 years from history and is already exceeded at the Advice-rule start; Operating Model B_{MSY} is not reached in 20 years from either start.

ing the same qualitative pattern holds, but the gain is small and Future still never reaches ICES $MSY B_{\text{trigger}}$ (Section 3.3): cutting F to the HCR is not sufficient, under this SRR and mean recruitment, to satisfy the ICES threshold. For mackerel, Historical $SSB/MSY B_{\text{trigger}}$ does not move; the HCR trims F because the stock is below the SAG trigger, while Operating Model B_{MSY} is already exceeded.

Irish Sea cod is the implementation contrast in the other direction. Realised F is far below the single-stock target, and imposing F_{target} or the HCR reduces terminal SSB. Calling that an implementation failure would be a category error: catch sat below what a single-stock Category 1 rule would have allowed. Whether that was a mixed-fishery choke, a recovery TAC, or a reference-point mismatch is not identified here. Until an advised-catch trajectory is run, implementation and the advice rule remain only partly separated (Punt et al., 2016; Winker et al., 2026).

4.2 Mackerel reference-point basis

Northeast Atlantic mackerel is retained, but its $MSY B_{\text{trigger}}$ ratios are not on the same published basis as the gadoids. The SAG $MSY B_{\text{trigger}}$ used as an HCR input equals B_{pa} in the latest SAG row; the advice sheet uses a long-term management plan / B_{lim} convention. A recent benchmark shifted both the stock trend and the reference-point basis. The result that mackerel is already above Operating Model B_{MSY} while still below the SAG trigger is therefore a statement about those two numbers on this Operating Model, not a claim that ICES advice and MSY biomass agree. Future rebuild times to $MSY B_{\text{trigger}}$ (3 years) should be read with that caveat.

4.3 Structural uncertainty forthcoming

SRR form is a discrete structural-uncertainty axis, not a process-error replicate. This paper reports bh3 on both Historical and Future; ICES HCR parameters remain the published SAG values on eqs. Figure lines for B_{lim} , $MSY B_{\text{trigger}}$ and F_{target} are those ICES inputs; Operating Model shading is bh3 . bh3 is Beverton–Holt with an R_0 prior from a hockey-stick; it is not a steepness-1 Beverton–Holt. bh1 is unconstrained bevholtsV with estimated steepness. Scenarios still to be reported are bh2 (FishLife steepness prior) and the Ricker analogues rk1 – rk3 . HCR inputs will remain at published ICES values across those scenarios; Operating Model B_{MSY} and F_{MSY} will be recalculated per scenario.

This draft is deterministic (residuals replayed once; Future at mean recruitment). It does not propagate estimation error, does not refit SAM, and does not use a shortcut assessment emulator. Those are subsequent labelled axes, not results claimed here.

Acknowledgements

This work was carried out at Sea++. Laurence Kell is grateful to colleagues in the ICES community whose assessments and advice sheets provide the Operating Models and reference points used

here. Errors of interpretation remain the author's.

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